18. Let $X$ have a Bernoulli distribution with mean 0.4, then the variance of $(2X–3)$ is

(1) 0.24 b) 0.48 (3) 0.6 (4) 0.96

19. If in 6 trials, $X$ is a binomial variable which follows the relation $9P(X=4) = P(X=2)$, then the probability of success is

(1)0.125 (2) 0.25 (3) 0.375 (4) 0.75

20. A computer salesperson knows from his past experience that he sells computers to one in every twenty customers who enter the showroom. What is the probability that he will sell a computer to exactly two of the next three customers?

(1) $\frac{57}{20^3}$ (2) $\frac{57}{20^2}$ (3) $\frac{19^3}{20^3}$ (4) $\frac{57}{20}$

## SUMMARY

- **A random variable** *$X$ is a function defined on a sample space $S$ into the real numbers $\mathbb{R}$ such that the inverse image of points or subset or interval of $\mathbb{R}$ is an event in $S$, for which probability is assigned.*
- A random variable $X$ is defined on a sample space $S$ into the real numbers $\mathbb{R}$ is called discrete random variable if the range of $X$ is countable, that is, it can assume only a finite or countably infinite number of values, where every value in the set $S$ has positive probability with total one.
- If $X$ is a discrete random variable with discrete values $x_1, x_2, x_3, \ldots x_n \ldots$, then the function denoted by $f(.)$ or $p(.)$ and defined by $f(x_k) = P(X = x_k)$ for $k = 1,2,3,\ldots n,\ldots$ is called the probability mass function of $X$
- The function $f(x)$ is a probability mass function if

(i) $f(x_k) \geq 0$ for $k = 1,2,3,\ldots n,\ldots$ and (ii) $\sum_k f(x_k) = 1$

- The **cumulative distribution function** $F(x)$ of a discrete random variable $X$, taking the values $x_1, x_2, x_3, \ldots$ such that $x_1 < x_2 < x_3 < \cdots$ with probability mass function $f(x_i)$ is

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i), \quad x \in \mathbb{R}$$

- Suppose $X$ is a discrete random variable taking the values $x_1, x_2, x_3, \ldots$ such that $x_1 < x_2 < x_3, \ldots$ and $F(x_i)$ is the distribution function. Then the probability mass function $f(x_i)$ is given by $f(x_i) = F(x_i) – F(x_{i–1})$, $i = 1,2,3, \ldots$
- *Let $S$ be a sample space and let a random variable $X : S \to R$ that takes any value in a set $I$ of $\mathbb{R}$. Then $X$ is called a* ***continuous random variable*** *if $P(X = x) = 0$ for every $x$ in $I$*
- *A non-negative real valued function $f(x)$ is said to be a* ***probability density function*** *if, for each possible outcome $x$, $x \in [a,b]$ of a continuous random variable $X$ having the property*

$$P(a \leq x \leq b) = \int_a^b f(x)\, dx$$



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